

5058 Homework 4

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1 Stationary ARMA processes

The given AR(2) process is defined by:

$$(1 - 1.1B + 0.18B^2)Y_t = \varepsilon_t$$

where B is the backward shift operator and ε_t is white noise with $E(\varepsilon_t \varepsilon_\tau) = \delta_{t\tau}$ (variance $\sigma_\varepsilon^2 = 1$).

1.1 (a) Covariance-stationarity analysis

To determine if the process is covariance-stationary, we examine the roots of the characteristic equation associated with the AR polynomial $\phi(B) = 1 - 1.1B + 0.18B^2$:

$$1 - 1.1z + 0.18z^2 = 0$$

Solving for z :

$$z = \frac{1.1 \pm \sqrt{1.21 - 0.72}}{0.36} = \frac{1.1 \pm 0.7}{0.36}$$

The roots are $z_1 = 5$ and $z_2 = \frac{10}{9} \approx 1.11$. Since both roots $|z_i| > 1$ lie outside the unit circle, the process is **covariance-stationary**.

1.2 (b) Autocovariance calculation

For an AR(2) process $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$, the Yule-Walker equations are:

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2}, \quad k \geq 1$$

Given $\phi_1 = 1.1$, $\phi_2 = -0.18$ and $\sigma_\varepsilon^2 = 1$:

$$1. \text{ For } k = 1: \gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1 \implies \gamma_1 = \frac{1.1}{1.18} \gamma_0 \approx 0.9322 \gamma_0$$

2. To find γ_0 , we use the relation for $k = 0$: $\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + 1$.

Substituting $\gamma_2 = \phi_1\gamma_1 + \phi_2\gamma_0$ into the equation:

$$\gamma_0 = \frac{(1 - \phi_2)}{(1 + \phi_2)((1 - \phi_2)^2 - \phi_1^2)} = \frac{1.18}{0.82(1.18^2 - 1.1^2)} \approx 7.89$$

Solving this yields the variance $\gamma_0 \approx 7.89$.

2 Linear regression models

2.1 (a) Unit root test (ADF Test)

(i) t-statistic and significance level

The t -statistic for the lagged term X_{t-1} is:

$$t = \frac{\hat{\beta}_{X_{t-1}}}{SE(\hat{\beta}_{X_{t-1}})} = \frac{-0.11}{0.04} = -2.75$$

Based on the ADF table for sample size $T \rightarrow \infty$ with a constant: The critical value for 10% significance is -2.57 , and for 5% is -2.86 . Since $-2.86 < -2.75 < -2.57$, we reject the null hypothesis at the **10% significance level**.

(ii) Conclusion

The rejection of the null hypothesis H_0 implies that the process X_t does not have a unit root and is **stationary**.

2.2 (b) Model Comparison and Forecasting

(i) t-statistic of Y_{t-1}

In model (3), the t -statistic for Y_{t-1} is:

$$t = \frac{-0.16}{0.07} \approx -2.286$$

For a large sample, $|-2.286| > 1.96$ (5% critical value). Thus, we reject H_0 at the **5% significance level**.

(ii) Forecast for X_5

Given: $X_4 = 1.4, X_3 = 2.9, X_2 = 4.3, X_1 = 0.8$ and $Y_4 = 7.0$. $\Delta X_4 = -1.5, \Delta X_3 = -1.4, \Delta X_2 = 3.5$.

Model (1): $\Delta X_5 = 0.002 - 0.31\Delta X_4 = 0.002 - 0.31(-1.5) = 0.467$

$$X_5 = 1.4 + 0.467 = \mathbf{1.867}$$

Model (2): $\Delta X_5 = 0.021 - 0.46\Delta X_4 = 0.021 - 0.46(-1.5) = 0.711$

$$X_5 = 1.4 + 0.711 = \mathbf{2.111}$$

Model (3): $\Delta X_5 = 1.279 - 0.51\Delta X_4 - 0.44\Delta X_3 - 0.30\Delta X_2 - 0.16Y_4$

$$\Delta X_5 = 1.279 + 0.765 + 0.616 - 1.05 - 1.12 = 0.49$$

$$X_5 = 1.4 + 0.49 = \mathbf{1.890}$$

Model	Predicted X_5
(1)	1.867
(2)	2.111
(3)	1.890

3 Fourier Transform of Impulses

In this problem, we use the definition of the Fourier Transform: $\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$.

3.1 (a) Gaussian Impulse

Given $f(t) = ce^{-b(t-t_0)^2}$. Using the standard integral $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi/a}$, we perform the transform:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} ce^{-b(t-t_0)^2} e^{-i\omega t} dt$$

Let $u = t - t_0$, then $t = u + t_0$:

$$\hat{f}(\omega) = ce^{-i\omega t_0} \int_{-\infty}^{\infty} e^{-bu^2 - i\omega u} du$$

Completing the square in the exponent: $-bu^2 - i\omega u = -b(u + \frac{i\omega}{2b})^2 - \frac{\omega^2}{4b}$.

$$\hat{f}(\omega) = ce^{-i\omega t_0} e^{-\frac{\omega^2}{4b}} \int_{-\infty}^{\infty} e^{-b(u + \frac{i\omega}{2b})^2} du = c\sqrt{\frac{\pi}{b}} e^{-i\omega t_0 - \frac{\omega^2}{4b}}$$

3.2 (b) Dirac Delta Impulse

Setting $b = 1/(2\sigma^2)$ and $c = 1/\sqrt{2\pi\sigma^2}$.

(i) Properties as $\sigma \rightarrow 0$

As $\sigma \rightarrow 0$:

- For $x \neq 0$, $f(x) \propto \frac{1}{\sigma} e^{-x^2/2\sigma^2} \rightarrow 0$ because the exponential decay dominates.
- For $x = 0$, $f(0) = \frac{1}{\sqrt{2\pi}\sigma} \rightarrow \infty$.
- The integral $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2} dx = 1$ (standard normal distribution total probability).

Thus, it satisfies the properties of $\delta(x)$.

(ii) Fourier Transform of Dirac Delta

Substitute b and c into the result from (a):

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}\sigma} \sqrt{2\pi\sigma^2} e^{-i\omega t_0 - \frac{\omega^2 \sigma^2}{2}} = e^{-i\omega t_0} e^{-\frac{\omega^2 \sigma^2}{2}}$$

As $\sigma \rightarrow 0$, $e^{-\frac{\omega^2 \sigma^2}{2}} \rightarrow 1$. Therefore, $\mathcal{F}[\delta(t - t_0)] = e^{-i\omega t_0}$.

3.3 (c) Step Function

(i) Fourier Transform

Using $f(t) = e^{-at}$ for $t > 0$ and taking $a \rightarrow 0$:

$$\hat{f}(\omega) = \lim_{a \rightarrow 0} \int_0^{\infty} e^{-at} e^{-i\omega t} dt = \lim_{a \rightarrow 0} \left[\frac{e^{-(a+i\omega)t}}{-(a+i\omega)} \right]_0^{\infty} = \lim_{a \rightarrow 0} \frac{1}{a+i\omega}$$

Using the Plemelj formula relation $\frac{1}{a+i\omega} = \frac{a}{a^2+\omega^2} - \frac{i\omega}{a^2+\omega^2}$, as $a \rightarrow 0$:

$$\hat{f}(\omega) = \pi\delta(\omega) + \frac{1}{i\omega}$$

(ii) Power Spectral Distribution

The power spectrum is $S(\omega) = |\hat{f}(\omega)|^2$. For the step function, the distribution is dominated by the $1/\omega^2$ decay, representing a signal with high energy at low frequencies.

3.4 (d) Sign Function

$sgn(t) = f_{step}(t) - f_{step}(-t)$. Using the linearity of Fourier Transform:

$$\mathcal{F}[sgn(t)] = \left(\pi\delta(\omega) + \frac{1}{i\omega} \right) - \left(\pi\delta(-\omega) + \frac{1}{-i\omega} \right) = \frac{2}{i\omega}$$

3.5 (e) Power Law Decay Transforms

Using the duality property $\mathcal{F}[\mathcal{F}[f(t)]] = 2\pi f(-\omega)$:

1. $f(t) = 1/t$: From (d), $\mathcal{F}[sgn(t)] = 2/i\omega$. By duality, $\mathcal{F}[1/t] = -i\pi sgn(\omega)$.
2. $f(t) = |t|$: Since $\frac{d}{dt}|t| = sgn(t)$, using $\mathcal{F}[f'] = i\omega\hat{f}$, we get $\mathcal{F}[|t|] = -2/\omega^2$.
3. $f(t) = 1/t^2$: Using the derivative property on $1/t$, we get $\mathcal{F}[1/t^2] = -\pi|\omega|$.